

ActInf GuestStream #015.2 ~ Bobby Azarian "The Integrated Evolutionary Synthesis"

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hello everyone this is the active inference institute it is guest stream number 15.2 on july 26 2022 we're here with bobby zarian and this is going to be the second part of a discussion that we opened some months ago and uh looking forward to it really appreciate your visit and please just set it up and continue however you see fit i'll be documenting questions in the live chat and looking forward a lot to this discussion okay thanks for having me so uh the talk isn't quite uh what i promised it's not part two i'd like to come back in the future and talk about you know that stuff um it's kind of uh what i talked about last time but a little bit more focused uh more specific there was a recent guardian article that came out that was talking about whether there's this need for a new evolutionary synthesis we have the modern synthesis and then the modern synthesis has been challenged by the extended evolutionary synthesis which includes more mechanisms things that weren't recognized by the modern synthesis so things like epigenetics and lamarckian mechanisms like the baldwin effect uh multi-level selection or group selection um so this is actually saying that that's not radical enough uh the integrated evolutionary synthesis uh has that name because it integrates evolutionary theory with uh thermodynamics and information theory and give gives us um i think a much clearer picture of how evolution works and why it's generating complexity in the biosphere so originally this talk was also titled something different it said a unifying theory of reality and yeah it's been kind of downgraded to something a little less ambitious because i did not have time to include slides uh that talk about uh there's an interpretation of quantum mechanics called quantum darwinism and a solution to the fine-tuning problem called cosmological natural selection so what i'm trying to paint today is a picture of a darwinian reality and those uh paradigms are consistent with this darwinian framework uh kind of a universal darwinism type thing uh danned in it you know comes to mind um so this is just gonna focus on uh the evolution of the biosphere the evolution of life towards higher complexity and in the future we can talk about all that other stuff awesome sounds good yeah um who would you say this is for and how

would you like to update their cognitive model yeah so that's i love that you put it that way yeah i hope i'm updating people's model of reality uh today so uh it's for the active inference lab so it's a bit more technical than you know it might be if i was giving it to a general audience but i tried to not get too technical at the same time in my last talk i really just talked about everything and um used a lot of terminology that uh i assumed that your audience would know today if anything's unclear uh or you want um uh me to go into a little bit more detail uh please feel free to ask jump in i'll try to pause um a few times throughout the presentation and see if anybody has any questions because it's really this story of evolution that's something that you know you can tell as a story um and um you know i want people people to understand the mechanisms better than giving like the perfect presentation i just want people to see how all of these things work together um so it's from my book that just came out a month ago and that's available so if you like what you hear this is basically a kind of superficial uh version of the content in part two of the book there's three parts so there's this uh kind of famous quote from dobzhansky who did a lot of work uh creating the modern synthesis and the modern synthesis is basically um taking darwin's theory of natural selection and updating it with a modern understanding of genetics so it's not really modern anymore i mean this was like you know the gene was uh dna was discovered in the 50s and then in the 60s and 70s uh this was uh formulated so he said nothing in biology makes sense in except in the light of evolution i think it's a great statement i think it's true but the integrated evolutionary synthesis uh says that doesn't go far enough that nothing in biology or evolution makes sense except in light of thermodynamics and information and shannon's information theory is really the same theory as statistical thermodynamics there's a lot of the same formalisms and we're gonna see today that um biological information really the role of information uh is that it allows life to sustain itself against this tendency toward decay described by the second law of and basically there's just one evolutionary algorithm one mechanism that is behind all this complexity creation and that has been called variation and selection uh that's a term from evolutionary epistemology which is based on the ideas of karl popper but it actually wasn't founded by karl popper it was founded by a cognitive psychologist named donald campbell who was inspired by popper's view and then popper contributed to it later but the idea is that variation and selection the mechanism of evolution is a form of trial and error learning so evolution adaptation is the same as learning it's a learning mechanism and it creates biological knowledge it creates knowledge in the biosphere um and david deutsch uh is one person who has written a lot about uh evolution as a knowledge creation process and i think how we need to understand it from

now on and this is arguing that evolution does have an arrow or a direction it is progressive in other words [Music] in the 20th century stephen j Gould was very adamant about killing this idea that evolution is progressive uh he you know saw it as being teleological um kind of implying that there's some mystical force driving uh evolution towards you know higher levels of complexity and i think that is wrong um evolution is progressive but there's no mystical force driving it uh it is just mechanisms that are completely uh intelligible and uh formalizable and richard dawkins was one person who uh challenged gold on that he wrote a book review uh about gold's book and basically said that this idea that evolution towards higher complexity was this statistical illusion was wrong and he cited uh evolutionary transitions as being a mechanism of complex degeneration and we're going to talk about uh evolutionary transitions today which is left out of the modern synthesis they don't talk about these transitions this uh quote from stuart kaufman he says there are things darwin couldn't have known one of them was self-organization in complex dynamical systems if the new science of complexity succeeds it will broker a marriage between self-organization and selection it'll be a physics of biology so that's part of what we're going to talk about today uh how self-organization fits into the evolutionary picture because it wasn't recognized by most evolutionary biologists for a long time and santa fe institute was one place really you know doing the complexity research that that has made you know this mechanism of uh kind of famous um it started off with uh the cyberneticist so ross ashby had a principle of self-organization in i think the 40s or 50s so it's an old idea and it's you know time that uh i guess mainstream evolutionary theory incorporates self-organization into the picture and we'll see that self-organization is really the process that creates the evolutionary transitions that i mentioned so we're going to start off with some quotes just to give some context before getting into the details thomas huxley a 19th century english biologist known as darwin's bulldog for his advocacy of evolutionary theory said that the question of all questions for mankind the problem which underlies all others is more deeply interesting than any other is the ascertainment uh and of the place which man occupies in nature and of his relations to the universe of things so what's our relationship to the cosmos do we have one is life completely insignificant or could it have deep cosmic significance could it influence the future evolution and developmental trajectory of the universe so here's one answer to that question the kind of pessimistic answer man at last knows that he is alone in the unfeeling immensity of the universe out of which he emerged only by chance so this quote comes from jacques mono a nobel prize winning french biochemist a really smart guy this book was really influential and kind of painted this nihilistic

picture of reality that was consistent with the reductionist model of the world which was you know the reigning view in the 20th century and today we're going to challenge that idea uh christian de duve another nobel prize winning biologist uh said that the natural constraints within which chance operates are such that evolution in the direction of increasing complexity was virtually bound to take place if given the opportunity chance does not exclude inevitability so there's certain certainly chance and contingency and randomness operating in evolutionary processes you have genetic mutation which is at least partially a random blind process and uh that's okay that doesn't mean the whole thing is random so there's a lot of evidence from convergent evolution the work of simon conway morris that uh shows that uh the certain you know designs organism designs are regularities so we see them uh emerge kind of expectedly david deutsch uh founder of quantum computation and author of some great books fabric of reality and his newer the beginning of infinity said narrowly conceived evolutionary theory considers as mere vehicles for the replication of our genes or memes and it refuses to address the question of why evolution has tended to create ever greater adaptive complexity or the role that such complexity plays in the wider scheme of things so you see right here he was hinting at life which he calls adaptive complexity uh as having this larger cosmic role and i think adaptive complexity is a great term for life uh kind of replacing the word life that would be a good idea because life has these mystical connotations so soon as you say something like life is in some sense some statistical sense destined to spread through the universe it sounds mystical but when you say adaptive complexity uh will spread um you get a lot less groans because uh it makes it clear that what life is is this phenomenon that learns living systems are complex adaptive systems so adaptive complexity is just a way to talk about the network of complex adaptive systems that make up the biosphere and once you learn that life is really this phenomenon that is continuously learning and self-correcting then you see why uh it's so robust and why it seems like there is this inevitability to evolution and life getting off of its planet of origin so we're going to kind of demystify that process but in demystifying it we see a natural process that is pretty mystical as far as like you know just i i think it doesn't make sense to to use the term like to not say that there's not this like kind of magical or mystical quality of the process just because we can understand it mechanistically to me this paints a picture of nature that's you know more wondrous than um any sort of conception of nature that involves like supernatural things i think the the natural self-organizing is as interesting as you know any supernatural concept and then you see this really uh bold quote by uh the great physicist freeman dyson who just passed away like a year or two

ago it is conceivable that life may have a larger role to play sorry i can't see the quote then we have yet imagined life may succeed against all odds and molding the universe to its own purpose so we're gonna see if um we can make an argument for that last quote paul davies says we now see how it is possible for the universe to increase both organization and entropy at the same time the optimistic and pessimistic arrows of time can coexist the universe can display creative unidirectional progress even in the face of the second law so nothing in this story uh in any way violates the second law of and it's important that you know that's emphasized because a lot of people had this just superficial interpretation of the second law that the universe is growing increasingly disordered on average over time this presentation will say that that um you know popular understanding uh of the second law is wrong and that the universe is becoming more organized uh at the largest scales okay let's get into it if anyone has any questions about that introduction nope okay so the paradigm of emergence cosmic evolution says each emergence in the cosmic self-organization process is brought together by a meta-system transition or an evolutionary transition that moves life further away from a state of thermodynamic equilibrium and total disorder through a nested series of such phase transitions where functional things come together to make larger functional things which come together to make even larger ones and so on adaptive complexity becomes better equipped to dominate the cosmos life acquires an organization that is increasingly hierarchical and integrated and therefore more resilient and computationally powerful so this is what that process looks like you see this structure of hierarchy and of matter which continues with the hierarchy of life uh so we're gonna show how these new levels emerge and on the right here you see that uh there's this hierarchy of science that has tried to you know model this natural phenomenon phenomena and you get these levels that represent these different emergencies so with biology we have life then we have the emergence of mind and we get psychology and then we have the emergence of civilization and we get culture this framework says that and it's really the framework of evolutionary epistemology and universal darwinism it says that the emergence and the evolution of life mines society and technology are all part of one thermodynamic process one evolutionary process one computational process unified by the concept of knowledge and it says that humans are neither a cosmic accident nor the end goal of humans are a step on the evolutionary ladder of becoming and an essential driver of increasing complexity and it says the universe is undergoing a grand and majestic self-organizing process and at this moment in time in this corner of the universe we are the stars of the show this is what this process of cosmic evolution looks like uh this comes from a book

called cosmic evolution by the harvard uh astrophysicist eric chason uh these are some unifying paradigms of the past and present so the integrated evolutionary synthesis is not really a novel thing it's sort of an update to all of these uh unifying paradigms that have come before it so general systems theory came out of cybernetics uh it was founded by ludwig von berdalani and basically says that all complex systems have similar organizing principles that can be described conceptually and modeled mathematically the problem with this theory was that it was a little too broad and not specific enough evolutionary epistemology came after that and it emphasizes that the evolutionary process is a problem-solving procedure that creates knowledge the growth of complexity and the emergence of intelligence in the biosphere arises out of a knowledge accumulation process in this view when i say knowledge accumulation uh this knowledge is encoded in uh the different forms of memory um that life allows for so genetic memory neural memory uh cultural memory and we'll talk about that more later universal darwinism came after evolutionary epistemology was basically the same theory though so richard dawkins and dan din are people that are associated with universal darwinism and it emphasizes that the universe is evolving at all these different scales and levels of complexity so universal bayesianism is a term that some people are using and it's just basically an update of the previous paradigms so it basically says the same things that you know now evolution is the knowledge creation process but it specifically says that this knowledge is encoded in biological systems uh that knowledge is stored in the form of a world model and that the model is an internal representation and statistical mapping of the environment which gets updated through adaptation and learning and we're going to see that adaptation is a form of learning and that learning is a form of adaptation so to kind of frame this story earlier i said it's you you can tell the story in a kind of like mythological way so david deutsch says if one can speak of a cosmic war it's a war between monotony and novelty between stasis and creativity and in this war our side is not destined to lose if we choose to apply our unique capacity to create explanatory knowledge we could win so it's a pretty uh optimistic quote um and a story that i think you know is probably a lot more comforting satisfying than the old view that the universe is becoming increasingly disordered and that life is just a transient uh phenomenon we'll see that adaptive complexity uh isn't transient you could frame this as this battle between order and disorder or life versus entropy or knowledge versus ignorance and so the story starts with the second law of thermodynamics which is kind of a nice way to frame the evolutionary story since the second law of thermodynamics is this kind of iron-clad law that everybody accepts so it's kind of built on a really uh firm theoretical foundation so to

understand life and its role in cosmic evolution we have to understand the second law which says that the entropy of a system naturally increases over time at least on average you could sum this up by quote things fall apart so it's describing this tendency toward decay and disorder and we see that in this little simple cartoon below so in our everyday experience things don't typically organize themselves rooms get messier and buildings erode not the opposite but okay so yeah this is uh just a depiction of that same process uh which was the model that the founder of the statistical interpretation of the second law uh used so ludwig boltzmann imagined a simple system so an ideal gas in a box and you see that if this system is just left to evolve over time uh the these molecules spread out simply because uh if they're just all moving at random um then there are many more configurations that the system can be in that are disordered compared to patterned arrangements so just due to this statistical tendency uh you will get a system evolving toward a more disordered state if it's a closed system and there's no way for the system to access any energy from the outside so this is you know this model is looking at a gas in a box it's not looking at the universe but people were quick to jump uh to apply this to the universe as a whole under the assumption that the universe is a closed system uh but it ignores a lot of things so this model ignored the effect of gravity um because gravity doesn't really have a noticeable effect on things that small so it was just seen as negligible and it also assumes that these particles stay uncorrelated after they interact we know that with you know certain chemical systems in life the molecules are had held together by chemical bonds and forces this model ignores a lot of things which make it probably not very useful to apply to the universe as a whole and it leaves out this phenomenon of life that we've called adaptive so if uh systems naturally become more disordered than what's going on with life we see life maintaining its structure and you know re has the capability of regenerating itself and making copies of itself and producing all of this order that we see around us in the biosphere so adaptive complexity is this local reversal of the second law and schrodinger was one of the first people to really talk about this mystery give us one piece of the puzzle so life is able to evade this tendency towards disorder because it is an open system the planet is receiving all of this free energy from uh the sun and that energy uh allows us an adaptive system to stay far from thermodynamic equilibrium so the trick to life evading the tendency toward decay is that it has the ability to extract energy that allows it to do work and that work is maintaining its uh stable non-equilibrium steady state and uh so you can think of life as playing a game so the game of life uh is this battle between order and disorder so the challenge that life has that any system any adaptive system any conceivable sentient system in

any universe abiding by the second law has this intrinsic problem and that's it has to be able to resist the natural tendency toward disorder which requires work so this intrinsic challenge gives life an intrinsic goal life does have a goal and the goal is simply to continue to persist to stay far from thermodynamic equilibrium or to survive now what must life do in order to achieve that goal it has to be able to extract energy from the environment and that's not a trivial task in order to be able to navigate its environment and actually extract or absorb this energy uh it has to acquire information about the environment that it's in it has to model the world and um life strategy is to search the space of possible configurations for structural and functional solutions we call those adaptations to the problem of survival and so evolution is this process of life exploring the design space and uh dennit made this way of looking at it popular in his book darwin's dangerous idea so on an abstract level we can imagine a certain practical problem or challenge so any challenge we can imagine a challenge as having a solution that exists somewhere out there in the space of possibilities just waiting to be found by adaptive complexity so it may take some time but if the possibility space the space of possible solutions is continuously explored in an efficient way eventually a solution will be discovered so darwinian evolution is a problem-solving algorithm and that algorithm is called variation and selection um yeah this slide was old i meant to update that so this is supposed to show that variation and selection is the method of adaptation and that uh this is a trial and error learning process so trial and error is the mechanism for learning those are the same thing so adaptation is equivalent to problem solving is equivalent to learning is equivalent to knowledge creation and we're going to see a illustration of that to make it a lot clearer an evolutionary epistemology inspired picture that uh david deutsch supports so he says that the whole scientific process resembles biological evolution a problem is like an ecological niche and a theory is like a gene or a species which is being tested for viability in that niche the way we should think about organisms now is that they are embodied theories about how to survive in a particular niche so a genome is basically a strategy for survival okay so this picture looks complicated at first but it's pretty simple once you break it down so here on the left we see you know one of the most simple organisms we have a single-celled organism and this says that an adaptive system is an embodied theory about how to stay far from equilibrium and that basically it's searching for a winning design so a winning design is specifically one that allows for energy extraction from the environment so now we're just really understanding life as having to solve a thermodynamic problem it has to find free energy to maintain its non-equilibrium steady state and it does this without any consciousness or conscious intent at first

so let's imagine that this is the origin of life that we start with this one it's a replicator so it makes copies of itself and because genetic mutation is unavoidable the copying process will always make errors then you get a generation of new designs that are variations on the original design so here you see design one two three and four and you can have as many designs as copies that this organism can make so just through replication with mutation you get this variation and selection mechanism that will create learning so natural selection is a filter and natural selection filters out the dysfunctional designs we see that uh you could imagine that so a bacterium some bacteria perform a process known as chemotaxis so they will swim towards molecular food and away from toxins so they have some sort of very basic abstract world model of the environment some statistical mapping that allows the system to basically swim towards food and away from threats and you can imagine that this first one maybe it swims in random directions it doesn't swim towards food yet it just moves around randomly so it makes a copy of itself let's say that copy still has that same behavior and then you have another copy let's say this one because of random mutation swims toward poisons and away from food uh and this one third one just by chance uh has a a better design a more optimal design where it swims towards food and away from toxins and you can imagine you know however you know many types of different designs with different functions and so some of these designs will die out because they can't fulfill this task of extracting energy so they can't stay far from equilibrium they get filtered out and this winning design gets to make copies of itself and so this winning design is the one that can best predict its environment so it's the one that's best at extracting energy because it has the ability to predict its environment this process continues with a second iteration and it creates this new generation of designs again some of these designs will be dysfunctional or at least not optimal especially if they're competing uh dysfunctional ones will die out so natural selection acts as a filter again and again we have another generation so this iterative process is how basically the species design gets optimized and it is also a knowledge creation because this organism has a genetic code codes for for these functional designs or dysfunctional designs and random is creating random bits of code it's creating like random changes and some of that code will produce functional designs so while it's producing all this information some of it is information that's not predictive of the environment it's useless information the information that survives the filter of natural has proved itself to be knowledge and the reason i call the information in these systems that survive knowledge is because they contain information that reduces the organism's uncertainty regarding the environment so you can look at this whole process as basically a world model

and there's variations being created to this you know embodied uh internal model and that with these iterations the model is getting updated be more accurate so i'm not going to go into active inference and free energy principle and the bayesian brain hypothesis here but i will mention some things for people who already are familiar with it since this is the active inference channel that this whole process would be minimizing free energy or prediction error so as adaptation occurs and you have these different generations the model is getting updated in a bayesian fashion john campbell has done work showing that the evolutionary process is a process of learning and specifically bayesian inference and carl fristen has worked with him on that so i'm calling this knowledge the information that encodes these uh structures knowledge that comes from evolutionary so what do i mean by that so a dolphins streamline design which is a product of the information stored in its genome contains a knowledge of hydrodynamics and similarly an eagle's wing design contains a knowledge of aerodynamics so not only can we be certain that engineers see knowledge in these functional structures there should be little doubt that these designs provided the original inspiration for our aircraft and submarine technology yeah you can think of evolution as a knowledge creation process you can think about all the species that exist as being solutions to different types of environmental challenges humans scientists studying these species can learn a lot about how better manipulate the universe and better solve our challenges by looking at how evolution has solved challenges in the past this process of adaptation that's creating this knowledge is also creating statistical correlation between the organism and its environment when you understand this you see that natural selection is something like an information channel you could think of as pumping in information from the environment into uh life models its environment it's isomorphic to its environment so an organism is a model of its environment in the way that a key is a model of the lock it opens so um as this process that we see here takes place and you have these different iterations of the adaptation process it's becoming more correlated with its so we talked about these different designs and being different types of bacteria some that swim towards poisons and some that swim away and towards food and so we see this is the work of physicist carlo rivelli who is now interested in evolution and how evolution creates um his work builds on the work of david wolpert and artem kochinski at the santa fe institute just to explain this concept of adaptation creating statistical correlation and mutual information and this is really what underlies uh its ability to predict uh we have this simple example so a well-adapted organism is one that is statistically correlated with its so a bacterium that swims to the left when nutrients are on the left and swims to the right when nutrients are on the right

prosper a bacterium that swims at random has less chances so increasing statistical correlation represents the creation of mutual information and that comes from shannon's uh information theory so basically when you have this correlation basically the information in these organisms is predictive of the environment and this uh quote from fred dretzki who did a lot of the pioneering work applying information theory to uh this is from the biologist john maynard smith's essay on biological information dretzki argues as follows if some variable a is correlated with the second variable b then we can say that b carries information about a for example if the occurrence of rain is correlated with a particular cloud then the type of cloud tells us whether it would rain so we see here that adaptation equals statistical correlation equals mutual equals model optimization equals knowledge creation let me just skip some of this stuff so we see how knowledge is created uh in this example and species are becoming optimized for extracting energy in their niche but they're not becoming more complex per se so um since is an embodied theory about how to extract energy from the environment it kind of follows uh ocam's razor so uh it's really trying to be the simplest solution to a thermodynamic at the beginning of this talk i i said that this was going to explain mechanisms of complexity generation and this tendency towards higher intelligence that we've seen uh in our biosphere um but it's not the case that is creating uh is making every species more complex or more intelligent and stephen j Gould was right about that uh if uh organism is extracting energy and avoiding threats just fine in its niche um it doesn't need to become more complex so crocodiles and sharks are common examples of um organisms that haven't become more complex over millions of years of and there are examples like cave fish that um fish that you know weren't previously existing in caves uh got trapped in caves or some sort of underground environment where having eyes was no longer a selective advantage you saw those organisms actually become more simple if evolution isn't creating is it making organisms more complex and more intelligent are certain species then what how do we argue that intelligence is inevitable like how does that work it's not that evolution makes every species more complex it's that evolution will create a new species and there will be this growing ladder of complexity such that the new species that emerge have to solve more complicated energy extraction problems and that is uh basically what ratchets up complexity so we're going to talk about that now this tendency towards intelligence so we can think of each niche on earth as a sort of energy slot a thermodynamic slot for a given species and an evolving population of organisms efficiently searching the space will discover a solution to a thermodynamic problem that it didn't know existed so the discovery of a novel energy source

or energy extraction technique is both how a new niche and a new species come into existence so this darwinian evolution process which is also called phylogenetic learning this iterative learning um process that's its generational generational learning um will naturally lead to speciation because organisms will stumble upon ways to exploit thermodynamic niches that were previously inaccessible to life purely for design reasons let's look at that picture one more time when this organism self-replicates it's making these different designs some of them will be better at extracting that source of energy for the first organisms it was thought to be the geochemic geochemical energy uh that was around these hydrothermal vents these deep sea volcanoes and um you can imagine just by chance uh that it creates a copy of itself and that it discovers a configuration that can suddenly absorb sunlight so that's how we get the generation of photosynthetic bacteria and when you get uh this you know new source of energy unlocked then uh that bit is basically what creates a new species you don't just have this you know one species exploring this solution space you have a growing uh problem space so you will have uh organisms discovering new sources of energy that weren't available so basically if life is something like a game and it's trying to you know find these configurations that are best at predicting the environment it'll kind of just by chance you know discover that there's like a different aspect of the game so there it will unlock a new uh source of free energy uh which creates a new and a new niche okay so just to make that idea a little more clear there's a great paper called the energy expansions of evolution by olivia judson and she says the history of the life earth system can be divided into five energetic epochs each featuring the evolution of life forms that can exploit a new source of energy these sources are geochemical energy sunlight oxygen flesh and fire the first two are present at the start but oxygen flesh and fire are all consequences of evolutionary events since no category of energy source has disappeared this has over time resulted in an expanding realm of the sources of energy available to living systems and a concomitant increase in the diversity and complexity of ecosystems so why do increasingly intelligent species emerge so first we have these organisms around the hydrothermal vents i mentioned reductive chemoautotrophs there are solutions to the problem of how to extract energy from geochemical gradients uh photosynthetic bacteria the single-celled ancestors of plants are a solution to the problem of how to extract work from all the solar energy that was flowing through the planetary system and heterotrophic organisms then emerge which are organisms that eat other organisms and their solution to the problem of how to extract energy from life itself and this explains how intelligence emerges because once life starts having to model other agents with causal power and adaptive behavior the

computational task of extracting free energy gets increasingly difficult as increasingly complex species arise by virtue of having to model each other the complexity of some species gets ratcheted up by what's known as an evolutionary arms race here i'll mention this law from cybernetics created by ross ashby called the law of requisite variety and applied to evolution it basically an adaptive system must have as many states like cognitive states behavioral states as there are challenges in its environment and so when you have this process of new species emerging uh that are able that basically discover configurations that allows them to extract a new source of energy or eat a new source of energy you get this pressure towards or this tendency towards species have a larger repertoire of cognitive states and so they can deal with more challenges and so this increases uh an agent's empowerment that's a term from the computational neuroscience literature um harvard ai researcher alex wisner gross has this concept of a causal entropic force but basically we can see that uh the biosphere as a whole is increasing uh its repertoire of internal states and the most complex species is also um possessing a repertoire of inter of larger internal states and this is one explanation for this tendency towards higher intelligence just to reiterate those points once more why has the evolutionary record shown a trend of the emergence of increasingly complex forms when all conceivable niches on the earth were filled that wasn't the end for complexification new species create new niches because the free energy slot they provide is themselves so they become food for any sort of other organism that through blind variation and natural discovers a configuration that allows them to exploit that new source of food so the cyberneticist francis heiligin said it is well documented by evolutionary biologists that ecosystems tend to become more complex a number of different species increases and the number of dependencies in other languages linkages between species increases yale wilson said not only do ecosystems contain typically lots of niches that will eventually be filled by new species there is a self-reinforcing tendency to create new niches and stu kaufman uh actually has shown this parallel between uh niche emergence and the auto catalytic process that we see um with basically the precursor to life was this chemical auto catalytic set so stu kaufman says ecosystems are auto catalytic auto catalytically closed self-sustaining reaction networks that reliably drive up biological diversity and complexity as they self-amplify and evolve okay so that tells us how we get species of increasing intelligence but it doesn't answer everything about how that happens because it isn't telling us you go from single-celled organisms to multicellular organisms and it's not addressing the complexity that we see with these collectives with society so how do we go from multi-cellular organisms to these larger societies we've seen that

because of the internet uh that human societies have become integrated into this larger computational network that some have called a global brain because humans are basically nodes exchanging information in the same way um neuronal populations or neurons are exchanging information in the brain we need to talk about more than normal uh darwinian evolution competitive and uh we need to include self-organization in this model so before i start talking about self-organization and that will kind of be the second half of this and it's a bit shorter any questions about any of the mechanisms that have been presented so far yeah there's definitely a few questions would you like me to ask them now or do you think it makes more sense to just present everything you'd like to get out and then we can have just a full q a uh if the questions are like broad questions that aren't um specifically asking about like the mechanisms that were just described then we could wait but if it's about this stuff maybe it'd be good to talk about let's um go through the whole presentation and again i'll have um compiled questions by then thank you okay great all right so this is that same thing that we were looking at earlier um it was an older version looks a bit different of these arrows but yeah this is showing the same process this uh phylogenetic learning process that happens through competitive evolution so this is you know normal darwinian evolution but you see here at the bottom this part at the bottom right is showing that these organisms down here these different designs they don't have to compete they can cooperate they can find a synergistic collective configuration and form a larger unit this is the process of self-organization right here and basically that creates a new level of complexity so if you think of life as a game and it's this game where life's trying to you know life's exploring the design space looking for configurations that solve its thermodynamic problem of staying far from equilibrium this process of self-organization is how life graduates to a new level in the game so you don't always have to compete you can cooperate we're going to talk about self-organization as a darwinian process this marriage of self-organization and selection that stu kaufman uh talked about in that quote at the beginning so richard dawkins says darwin's survival of the fittest is really a special case of a more general law of survival of the stable and this concept of evolution being about survival of the stable will allow us to understand how self-organization is a darwinian process and by a darwinian process i mean that it works through the evolutionary algorithm that we've called here you see the self-organization process up top there's just a couple examples of collectives of higher level systems that are made of many units so it can be a collection of molecules forming something like an auto catalytic set or it can be a collection of agents it can be people in a society but basically this is the evolutionary

process it works through variation and selection because uh to have variation on a design or structure um you don't need replication you don't need a system that's making copies of itself that makes the process uh more efficient in ways but you can have a single system that's exploring a variety of internal and so you see this flow chart here where it says system blindly explores various configurations in the space of possible designs via trial and error so it's just being pushed by a flow of with humans we're powered by metabolism it's still a flow of energy pushing us technically and uh it's basically causing the system to randomly explore different states in the configuration space now those configurations that it discovers that allows the system to extract sufficient energy to stay out of equilibrium are the ones that get retained and the system is able to persist for longer it's the same variation and selection mechanism but here since there's only one system donald campbell the inventor of being influenced by the cyber cyberneticists called this blind variation and selective retention so the stable states are selectively selectively retained so those states that the system moves into that does not allow the system to persist uh are filtered out so really you can explain evolution with this simple kind of tautological phrase uh what works persists and it that dawkins um simplification of darwin's law as uh really being about the shows us why this integrated evolutionary synthesis is important we can see the second law's tendency toward disorder as being a selection pressure it's the second law that filters out the unstable designs and they're not just filtering out the unstable designs it's specifically filtering out designs that are less functional so the stable designs because of this constraint this thermodynamic constraint that that systems must be able to extract energy to maintain their order it is really discovering the most functional designs the designs that are best at predicting the and the designs that are uh most robust and most functional this process explains the origin of life so you have all these interacting components these are molecules the that are involved in organic chemistry um jeremy england who did this work when he was at mit named this process dissipative adaptation so he was showing how basically a flow of energy pushing a system of molecules far from thermodynamic equilibrium will lead to this increasingly ordered state because the system is exploring configurations finding those information getting stuck basically in the configurations that allow the system to extract and dissipate more energy discusses this principle of recursive it was the name was inspired by ross ashby's principle self-organization but this is basically explaining why interacting agents will consistently link up to form stable holes and that's because working together allows the individual systems to extract more energy with less work so there's this effect that has been called synergy nature promotes cooperation

because it's thermodynamically beneficial for all parties and for that reason synergistic collective configurations will eventually be discovered by a mini component system that's exploring various states or configurations through the blind variation and selective retention mechanism so agents will compete until they discover that working together makes everyone's task easier and this is a lesson for humanity because um the the challenges the sustainability challenges which is an entropy problem can only be solved these global sustainability challenges can only be solved if we work together working together increases the computational power of the system so it's it's self-organization i guess i think was kind of ignored by they they didn't want to recognize anything that sounded like a teleological mechanism like something like kind of mystical and it was like how you know there's a second law of thermodynamics how can there be spontaneous self-organization well the answer was it's not exactly spontaneous it requires energy flowing through the system pushing the system far from equilibrium and forcing it to explore the configuration space and that's how we see that it's a darwinian process that you it explores these different configurations and the configurations that are most functional are the ones that uh survive and pass the the filter of natural selection here's just uh illustration showing that self-organization is something that can happen at all of these levels uh complex adaptive systems are nested systems they're systems made of smaller systems that are interacting they build on one another so you have single-celled organisms coming together to form this multi-cellular unit like a slime mold you know humans are multicellular organisms that come together and make societies uh you have ant colonies here which do the same and so these form a collective unit that function very much like the units that they're made of so you see that adaptive systems can exist in all these levels and these adaptive systems have similar dynamics and work through this these same variation and selection mechanisms and that's how they evolve toward increasing robustness and complexity and computational power i'm jeffrey west's book scale this great book about that so this principle of recursive self organization says that as long as a biosphere is creating a growing variety of complex adaptive systems some of those systems will interact to produce higher level complex adaptive systems will which will come together to form even larger complexes this process continues at higher and more computationally sophisticated levels as time marches on familiar examples of superorganisms include ant colonies termite mounds human societies and the ecosystems that make up the biosphere and all of these distributed networks process information and fight to maintain their organization in ways that are surprisingly similar to the organisms they're made of so it tells us that knowledge of the

dynamics and mechanisms at one level can be informative of isomorphic phenomena at other levels so it's kind of interesting here because you see that this theory has predictive power that no reductionist model has this you know reductionist model where it's trying to look trying to understand reality and predict reality by looking at this most base level such as the interactions of like basic particles and forces don't give us this predictive power that this uh paradigm does because it basically says that we can use adaptive systems we can look at those patterns at one level and then we can use that to understand things that might not be easy to see at another level so for example um and this we saw that paper with stu kaufman saying that ecosystems were auto catalytic sets setting auto catalytic sets can tell us about ecosystem dynamics and vice versa uh studying human brains can tell us about the emerging global brain so this is something that you know i won't go into here maybe in the next talk um but there's this question of if there is a global brain is a global mind emerging could the global mind even be like a conscious mind uh and i think that something at this paradigm uh some sort of predictive power that it has would say that for that mind to emerge the system might have to duplicate dynamics that you see underlying conscious brain states so not all brain states are conscious states when you go to sleep before you enter a dream there are periods where consciousness ceases to be but the brain state is still functioning physiologically it's keeping you alive so um to understand how a global mind emerges from a global brain uh we would we probably want to look at we probably want to look at like human brains and looking at this transition between unconscious to conscious states um and every level has novelty and so they're not going to be exactly the same new properties will emerge so it's not certain that these predictions will tell you everything about the higher level that hasn't emerged but it will give you a sort of outline of the dynamics that could be really useful so this rep process of recursive uh self-organization creates this hierarchical emergence that we saw at the beginning now we see exactly why it happens these things come together these units come together to make larger units because cooperating makes the thermodynamic task of staying far from easier uh you get the emergence of new properties as these higher level emerge uh higher levels emerge and so with brains which is left out of this but uh with the emergence of brains you have a system that can encode the causal consequences of its actions in real time that allows the system to create a variable for itself before brains an organism that does something like interacts with the environment there's no synaptic plasticity there's no mechanism for creating a memory of the effects on that environment in real time and so what brains allow is the system encode the consequences of its actions which gives it an understanding of itself it's

really the beginning of self-modeling something that's in addition to um this concept of intelligence being about like the the size of the repertoire of mental states which is increased by this niche emergence process that we discussed you also need to think about these levels of awareness that come from self-reference in the form of um so that would be for another talk where we talk about like how does consciousness emerge from this but yeah i just wanted to point out that these levels of awareness uh are another important factor uh of the emergence uh and growth of uh that we can't leave out and it's an emergent property so this process of multi-level and recursive emergence is creating more complex structures these hierarchical adaptive systems and these systems are these evolutionary transitions are also revolutions in information storage and processing machinery so we start out with genetic material then brains emerge and we have neural memory that sits on top of the genetic memory and then we have collective form and then we get cultural memory so memory that's shared among all the brains of the individuals in the system and then we get things like books and journals and then eventually technological memory digital memory so you can see this process uh is a process of knowledge accumulation and that the biosphere and technology the technosphere you could call it this is all one continuous process i didn't talk about it here but one of the big insights of karl popper's was that science and adaptive learning all work through this variation and selection mechanism so it's called trial and error learning in the developmental psychology literature you can understand science as competing theories that that works through like this darwinian process and the mechanism he used the name for that was called conjecture and refutation but it's it's all trial and error learning it's all variation and you have these different models and then these models get tested and natural selection weeds out the models that aren't predictive of the what does the integrated evolutionary synthesis show us this theory of knowledge creation it shows us that knowledge is power so this utor was just an acronym for unifying theory of reality uh shows us that knowledge is power is not just a hollow buzz phrase from the digital age it's true in the most fundamental way uncertainty reducing information is life's first and last weapon in the ongoing war with disorder and it infuses organic matter with control and causal power without knowledge life cannot exist for more than a moment much less colonize the galaxy and beyond this suggests that setting out on the path toward cosmic superiority is not a choice that intelligent agents like homo sapiens make upon careful reflection nor is it just some quirky ambition we stumbled onto by chance humanity's collective desire to transcend mortality and expand outward into space apparent from our current scientific and technological endeavors emerges not incidentally but as an

inevitable consequence of the fact that continual knowledge acquisition is a fundamental biological imperative the sun is our source of energy it's what allows us to stay far from intelligent species that are modeling the world with increasing sophistication will realize that there's this thermodynamic game clock that if life is to persist in the long term it has to leave the planet spacex like isn't elon musk's like creative idea this is an imperative that any sufficiently intelligent species uh will recognize has hopefully shown how natural selection is an information channel that pumps information from the inanimate world into life and through that process nature begins modeling itself so matter starts finding a configuration and that allows it to model the world around it that it's part of so it's encoding its own structure reality is encoding its own structure and the universe begins to wake up some of the other wording was you know life ascending to this you know cosmic superiority and being able to dominate the cosmos but you can similarly look at whole story not as a battle between life and entropy and order and disorder but that those things are complex complement complementary so they sort of need each other and they it's not the case that life is dominating the cosmos it's that the universe itself is self-organizing through this process um and that basically uh life is and agents are these kind of sensory modalities for the universe i think it's uh appropriate to to look at it that way i don't think there's anything wrong with it um so here's a quote from harold morwitz who was a founding member of the santa fe institute uh he was a professor of mine and it was a big inspiration on the book he says we start with observations and if evolving if the evolving cosmos has an observed direction rejecting that view is clearly non-empirical there need not necessarily be a knowable endpoint but there may be an arrow and so christoph koch one of the world's most respected neuroscientists wrote the rise of sentient life within times wide circuit was inevitable teohar deschardin is correct in his view that islands within the universe if not the whole cosmos are evolving toward ever greater complexity and self-knowledge the universe is a work in progress the evidence from cosmology biology and history is compelling so we're seeing a kind of change in sentiment and it's been brewing for a long time but that this idea this paradigm of the universe being self-organizing and becoming increasingly complex completely scientific and i think given what we know it is um it has predictive power too we can understand we can predict things about future emergencies so i would like to see this uh paradigm of emergence replace the uh reductionist paradigm where is this taking us so this is a kind of far out speculation but i think you know given the story told so far it's okay to go here um paul davies super respected uh theoretical physicist says many scientists have sorry i can't see that whoops there was stuff over the text many scientists have

speculated that as the timeline stretches toward infinity so an emerging super intelligence will become more and more godlike so that in the final stage the sorry i can't see that the mind will emerge with the sorry about that that's kind of anti-climactic for me to mess up here so that in the final stage the supermind will merge with the universe mind and cosmos will be one he goes on then the whole character of the universe including the emergence of its laws and the nature of its state sorry of its states become inextricably intertwined with mentality with its mindfulness what would that look like this is from where occurs while the singularity is near where it shows this trajectory where uh life creates technology and it merges with technology and you get the universe waking up what about heat death uh does does this process happen and then fade out because the universe is going towards thermodynamic equilibrium well people like stu kaufman and david deutsch have challenged the heat death uh scenario more and more physicists are with different you know alternative models some of these models basically say that the expanding universe um allows for a continual free energy supply i will get into that in another talk uh but yes so where is this heading um maybe something like this maybe something like a cosmic mind can't rule that out and uh yeah this comes from that book uh the almighty carl fristen was nice enough to give me a blurb for it if you email uh the romance of reality if you buy it from amazon and or wherever and send me a screenshot of the receipt with your address i'll send you a signed numbered book plate it's like a sticker that goes inside the book and yeah thank you so much for having me again nice thank you very much bobby let's take a deep breath and then to the question and answer yeah awesome wow a lot there so i'm just going to ask some questions that were asked in the live chat during the stream and also just some other ones i wrote down so i'd like to start with one of the first parts of the stream you read the dabjansky quote nothing in biology makes sense except in the light of evolution so famous and memeable quote yeah my question was what else is necessary and or sufficient for sense making in biology like is evolution the only light or what what are the lights that help us bring light to those issues so first part is about what else is necessary or sufficient and the second part is how should we move forward knowing um what is necessary to make light of well i guess it depends on how you define evolution because some people might be defining evolution very narrowly in terms of like the modern synthesis and someone might say we need to understand self-organization we need to understand emergence but with this framework all of those things are included under the umbrella of evolution so i'd say that you need evolution to understand the generation of complexity and intelligence you need these you know it needs to be expanded to include these ideas uh about um multi-level emergence

and uh this process where you have like self-modeling so we need to understand like how self-reference fits into this picture brains allow self-modeling and um you know first you have the emergence of consciousness which in this definition i would be talking about you know subjective awareness not necessarily like self-awareness or self-conscious um so self-awareness would be a higher level of self-reference so um those basically that to explain all of this stuff the evolutionary model would have to be this model of cosmic evolution where you talk about you know adaptation through darwinian competitive evolution but you also have this cooperative evolution with self-organization creating these new emergencies and these emergencies bringing about new properties new causal powers and higher levels of awareness but uh you know it's i don't want to be it's like reductionistic to be like is it all evolution so i think it's important a lot of trends today you're seeing like whiteheads process ontology coming back and this understanding of you know systems as processes rather than things i think that's enlightening i think that we need to talk about philosophy and metaphysics even um so for example and i talked about this in my last talk but we need to think about you know is hard determinism true or is reality a little bit more complex where reality fundamentally has this probabilistic element at the lowest level but then you know these higher level systems uh with you know these deterministic trajectories and you get uh yeah so so because the the model the hard determinism model that comes kind of from this laplacian model of reality this idea that there's just particles following these trajectories it has metaphysical uh and philosophical implications like we have no agency that there's just one future and it's completely determined and if we had a powerful enough computer we could even predict that turns out that that's wrong this new model uh allows for true agency and it allows for um organisms that emerge that have something like free will and that term opens up a different bag of worms um that's like some you know i'd like to talk about that in the next presentation but when i say free will i'm talking about biological agency and the fact that making decisions um we're not just particles following trajectories we're actually cybernetic control units that have emerged through these evolutionary processes that i explained this time um basically those processes that are pumping information into the system that information gives agents causal power such that you will see systems behave very differently than inanimate systems so an inanimate system like a rock only moves if it's pushed by a force but living systems because they metabolize and store energy they have self-generated motion and they're not predictable with classical mechanics you have to have some sort of higher level theory like fristen's bayesian brain hypothesis the free energy principle so you ca it's not that organisms

aren't predictable at all uh you can actually predict what they will do based on saying for example that they're going to behave in a way that minimizes their model's prediction error and that's very different than trying to calculate what the system's going to do based on these based on microphysics first of all you can't do it second of all i'm saying that that that idea is uh misguided to begin with because um this process where natural selection pumps information into these adaptive systems creates systems that have uh what people like sarah walker and paul davies have called informational control so it creates these cybernetic systems which use feedback mechanisms to stay far from equilibrium and we can't understand causality without understanding the role that information plays in biology evolution can explain a lot but it requires us to talk about thermodynamics and information and really what information allows systems to do is to continue to persist in the world against this tendency towards decay okay i've noted it down i have uh causal powers free will agency the role of information evolution energy thermodynamics for the part three let's uh leave a little footnote there i wanted to add a few kind of resonances that i saw with active inference like places where there was a nice plug-in or like a saddle point between the way that you framed it just in this nice presentation and in some of the ways that people are modeling in active inference today so you provided a dawkins quote with survival of the stable and that is kind of like a operational definition for the related claim survival of the fittest and people sometimes say well that's circular and survival of the stable almost says like it steps into that circularity it's like yes in fact it is that way and that's what our operational definition is repeated measurements is going to be what a thing is over a given time scale so there might be some spatial or temporal scale where a gradient or a boundary or a blanket does exist but that's going to be like an observer and a relational specific situation rather than like an all or none question about what kinds of things and processes exist so um just wanted to give that one and then i have a few more but what do you what do you think about that like where does this come in and how do you think that's different than how sometimes people have framed evolution and ecology in the past so yeah there is this criticism of the survival of the fittest being this kind of tautological hollow statement because what's you know what survives is what's fit but what's fit and you define it as what survives so you're not really getting to the bottom of it but this that's one thing that's one of the like i guess highlights of this integrated synthesis is we get an answer to that um and you're right saying what's stable you know it actually kind of embraces that thing like what what works persist or what's stable persists kind of has that same problem but the thermodynamic uh perspective gives it a specific answer it

says what is stable is what's able to extract energy from the environment so it's what's functional so we finally have an answer to like what fitness is that isn't tautological it's it's a system's ability to extract energy to stay far from equilibrium but also avoid threats because if something kills you it's going to rupture the markov blanket and you are going to collapse to equilibrium you're going to die so yeah the answer to like what fitness is is this ability to evade thermodynamic decay thank you let me go to the next one of these resonances because again i really liked how it was laid out and i guess how many paths we walked around some core points you mentioned learning and and seeing those in a new light like on part of a new continuum and so that's a lot like an active inference how there's similarity in how perception is modeled and in how slower and even slower processes of learning and memory happen and all of that is framed as a type of parameter finding whether that's what organisms are doing which is realism that's one perspective or whether it's just a way that we can model organisms that's another perspective instrumentalism so like when the perception of the ball crossing the visual field is happening is that perception of the location especially if we think from this generative uh approach like the bayesian is it perception yes is it learning the parameter like in a technical sense like machine learning of a parameter is it learning the parameter representing the location also yes so there's kind of a connection between learning and adaptation because data and observations coming in through the blanket are sometimes like sparse conflated related through causal mechanisms that aren't directly observed all these challenges with them evolution shapes systems ability for their for their sense making with the statistical regularities of the and that's been something that has been approached from people outside of the free energy principle active inference just um from different angles for example like collective behavior with the work of debra gordon and thinking about how collective behavior evolves to be fitting statistical regularities of different niches like four ant colonies like you mentioned and then also on the other hand this uh imperative to be fitting literally the statistical regularities of the niche leads directly to the embedded and the inactive all the four e's etc and then the kind of piece that ties that all together is you talked about the need to extract free energy like in a chemical in a dissipative situation you need to extract chemical free energy liberatable work and then in order to navigate the uncertainty at higher and higher levels more and more planning has to come into play and then that is like modeled with variational free energy in active inference which isn't exactly like that liberatable biochemical energy but it seems to be in people interpreted differently but they overlap in certain situations because the systems that are planning well are planning well to extract energy the systems that

are not planning well are not succeeding at extracting effective energy so like there is a relationship even though those variables are not formally the yes yeah very cool so yeah it's it's this story of free energy that the active inference paradigm is explaining but it becomes literal minimizing information theoretic free energy or predictive model error is exactly what allows the system to minimize actual thermodynamic free energy so if you look at the biosphere as a whole system you can look at individual agents individual organisms or these collectives maxwell ramstead and first in have done work showing that you know at these talking about complex adaptive systems that our societies these things are minimizing free energy as well um so yeah basically um you have this story this kind of neat little story where you can explain it all in terms of like different types of entropies life is trying to keep internal entropy low and to do that it needs to extract free energy um but by extracting free energy it's dissipating more of the world's free energy supply so it's keeping internal entropy low and increasing external entropy and it does that by minimizing its prediction error and that's minimizing shannon entropy or uncertainty or ignorance yes um i think what one kind of goal of this you know integrated synthesis is to basically it can incorporate the the free energy principle because uh to minimize um actual thermodynamic free energy to stay far from equilibrium you have to minimize your uncertainty of the environment and that's what you know bayesian inference is doing there's one other point i was going to make about that yeah well yeah so um i mean the free energy principle i think on its own you know it's just saying that systems will minimize uh you know the model's prediction error but it doesn't talk about like how you get like more complex species or anything it doesn't really tell us this whole story of evolution it just talks about minimizing free energy so uh what this story does is starts to get us think about uh thinking about niches and um that each niche uh represents this set of challenges and the set of computational problems and that um uh systems uh basically when you want to think about like why complexity well so minimizing free energy like trying to reduce surprise will keep you um far from equilibrium only for so long so if the environment changes then that species will die that species will get filtered out by natural selection so um there is a benefit to being curious there's a benefit to seeking out surprise and that's because um basically um if you can find a new niche to exploit then you don't have to compete with uh everyone else in the species or whatever group that is it's not just about minimizing surprise in in in many cases um you know a more complex species the most complex species especially like it's beneficial be curious and do epistemic foraging and try to you know imagine the space of possible configurations where society is headed so there's this real benefit uh to to seeking

out new knowledge that i think the free energy principle may kind of leave out or hint at thanks for sharing that just to add a few more points um so you mentioned daniel dennett's 1995 book darwin's dangerous idea and one of the really memorable parts in that book is this distinction between the cranes and the sky hooks and the images like a building is constructed and then we're going to see the artifact but not the process of how it was made and then ask was this likely constructed by a crane that put the last windows on the top or put the last piece of um you know building on top or was there a skyhook built kind of a castles in the sky and then built down the constructivist answer just a construction site but like a biological constructivist answer which is to say like a developmental perspective it's like well it was cranes all the way down and smaller cranes are basically constructing larger cranes if needed and then at some level there's like some um way for the seething of tiny tiny cranes somehow to facilitate larger cranes arising um so i was reminded of the cranes and skyhooks and and then as you just brought up in that light dance last answer and in some of the figures the design approach connects us with that kind of darwinian constructivism that dennett laid out but also you connected that to process ontology that in a sense this constructivism this niche constructivism and the way that the bottom of forces are leading to increasingly powerful large-scale organizations of systems like whether we call it life or adaptive complexity like you laid it out i'm seeing a sort of omega point direction towards instead of like finite competition curiosity driven niche for certain systems or maybe something involving other scales where like there's a rebalancing of the relevance of different mechanisms because we're seeing the adaptive complexity process playing out in a different way but that was what i was thinking and wondering how this framing influences how we act yeah so i think um uh it basically demystifies what looks like a teleological process and i still think it's okay to call it teleological because it is goal directed but it's not teleological the way some you know theologians have used it to mean that there's this mystical force or something like well it's really interesting it gets into like language issues because like uh henry bergson who uh coined that term elon vitale and tiohar de chardin who had a similar concept of radial energy and even going back to aristotle and his intellection um in tilos um these people claim to be talking about something natural even teal hard and uh so what we're seeing is that you can have this directional evolutionary uh without there being any mystical force so it's you know it's not the case where we're demystifying everything and then getting you know because we got rid of this teleological force we show that it doesn't exist this mystical immaterial force like pushing things toward progress uh by showing that doesn't exist we're not killing the argument that

evolution is progressive we're actually showing that this idea can exist in a completely naturalistic framework and to me it makes it more interesting but you do clearly see this statistical tendency towards higher intelligence and not only that evolution advances at a rate that allows life to leave the planet before its star dies and that's specifically the process creates consciousness and that um basically create this imperative in the uh most intelligent species of a biosphere such that they understand that long-term persistence requires getting off the planet so um it is the case that that a lot of species just evolved to be um as complex as their niche and if they're getting their energy uh just fine like bacteria there's no reason for there to be any pressure towards higher complexity but with humans for we start to construct our environment we construct niches we build all of this order around us and we create these new selection pressures and we create all of these new biologically relevant variables so there's a tendency where the most complex species uh basically how many variables they have to model is open-ended and anyone who is willing kind of go beyond the amount of things that most people model is going to have some sort of advantage let me make that a little bit more clear so someone in some very simple let's say small rural conservative town i'm not saying that in a bad way but um you can imagine a you know a culture with a simple lifestyle it might not be adaptive to be someone like roger penrose like some freak genius i think i read that like pinrose gets like lost like walking around like his office and stuff like he's just really absent-minded because his head's probably in the clouds thinking about mathematics he might not do well he might even be considered like the towns like crazy person in a small conservative town so if your niche is simple you don't have to have a complex uh world model and something simple like a world view like a religion can serve you just fine it might actually be better because it's the sense making lens that simplifies reality if you're motivated uh and you're someone who wants to uh take on you know this i guess you know modeling more variables in the world there will always be uh niches for you to exploit at places like academic universities and corporations where there are these niches that create this environment where there's actually this pressure towards more complex models and that means these there will be this evolutionary tendency towards organisms with a larger mental repertoire um so this niche emergence process where you get these increasingly complex species you can actually measure that increase using something like integrated information theories phi which measures basically the size of the mental repertoire and uh so that's what this niche emergence process is doing it's creating increasingly complex species that are more intelligent because they have a larger repertoire of accessible mental states but on top of that you have this other

component of intelligence which relates how the system is reflecting on itself so you get these emergence of higher levels awareness and i didn't talk about this but that's really what the prefrontal cortex in the brain does it's a module that unites these other computational modules which often do processing uh individually um there's this global workspace that emerges and basically um that's a meta-system transition as kind of lessons for society i think uh part of intelligence is these levels of awareness where you have consciousness and then you have self-consciousness or self-awareness that leads to different types of behavior but then there's another level that you might call meta awareness and it's realizing that you're a conscious agent in a network of conscious agents that form this higher level meta system that we call society so uh basically to uh and i hinted at this during the talk to overcome our existential challenges our sustainability challenges um ensuring that artificial intelligence isn't used in this way that destroys civilization all of these big problems require us coming together and the nations working cooperatively because it solving those problems requires the computational power of the full system so we need to cooperate to basically exploit this synergistic effect that occurs when you work with others rather than working alone the other another thing i guess i'd say right now about um kind of lessons for society is that complexity is really a function of two things you need a lot of parts um and the parts need to be highly connected the more connections between the parts and the more diversity there is among the components then the more complex that system is and the more computational power it can have that's basically why you can't have life emerge if you only have like all hydrogen elements are all oxygen elements like you can't just have like one or two types of atoms you need um this distribution of molecules because they play they function as something like a division of labor so at the society level a society full of exclusively made of engineers or made of doctors or made of artists like none of those societies would be functional if they had one type of person so you need a diversity among the component parts and you want a lot of connections between those parts so technology that connects us not just the internet but social media and blockchain technology anything that bypasses middlemen and is starting to make all these connections between the units those things are good they can also be bad as well we don't want to connect in a way where we become some sort of homogeneous hive mind i know when i say the global brain a lot of people express problems with that fears over becoming this kind of hive mind which i think china's government is like like basically trying to create and they have this philosophy of that the the citizens recognizes that they're part of an interdependent whole and they kind of use this to you know tell people like uh convince people that like surveillance and

all these other things like you know that take away freedoms is good because you're doing it for the greater good so what this paradigm says is that we want to become increasingly connected but we don't want to be homogeneous because we want diversity among the component parts and a system like that especially a system that doesn't allow criticism doesn't allow for a diversity of ideas a system that's homogeneous doesn't allow for cultural diversity so diversity and interconnection is good we want to come together we want to bring the nations together but we also want to keep the identities of the nations at the same time we want to preserve the cultures a brain is super integrated uh with you know 80 billion neurons and 10 000 connections between each of those no those neurons but it's also structured hierarchically and you have these different levels of modules and you have you know you have like the fusiform face area processing phases you have like the visual cortex doing visual stuff all this all these things are specialized but they're working together and i think that provides a model for optimizing society evolutionary principles can inform how we structure our governance systems our economic systems a few more questions if you'd like is that cool sure yeah all right so you've mentioned a few things about the trajectory towards integration what do you say to you see that where you see it up until you don't and where you don't see it so how do we really assess what is like what must happen or what probably happens given our really limited knowledge and how do we sort of balance this optimization across multiple points in a design space with this n equals one trajectory that our one body one spaceship earth all this sort of stuff is um limited by in scenario that we're in yeah so i mean we can't see you know what we don't know uh so i'm definitely not arguing that this process is inevitable uh in the way that would make our civilization incapable of failing um evolution towards higher progress occurs because life is always learning from its mistakes so it's failing it's failing constantly i didn't mention this principle in the book i call it popper's principle but it says problems create progress so it's a story of this you know this eternal thermodynamic challenge of staying far from equilibrium which requires um which requires also avoiding threats and just maintaining the stable state in the face of uh you know a chaotic and noisy world that uh yeah it's basically the challenges that we face that uh force us to find solutions um so it's not this straight march of progress it's not teleological in that way where there's like nothing stopping our failure we're actually going to fail over and over but because there's memory we learn from our mistakes um as long as you know that those failures are encoded in memory in some way you get this progressive process as far as like how do we handle like what we don't know um we have to actually engage in this process that uh i would say is very bayesian um

we have to um try to imagine all of the possible futures we have to map out all of the counterfactuals because that sort of allows us to see the possibility space and if we don't do this we're like sitting ducks like there's there's no way for us to be able to handle the challenges that are coming it was interesting like writing the book and thinking about these ideas and you know going from non-equilibrium thermodynamics to finding out about like the free energy principle and that it's part of this non-equilibrium story like how systems persist in a universe abiding by the second law of i was talking i was telling a friend about it um who was in design school i think it was parsons design school and they had this class they were teaching on future forecasting and it was literally i guess it was for people like architects all types of jobs but their job was to map out like the different possibilities of reality and like the different trajectories of ways we could go and they had to map out what was like likely and like less likely and you know at all these different levels of likelihood but if one thing if current events have showed us anything it's that don't always happen that we predict are likely i don't think trump becoming president would have seemed likely 10 years ago i don't think the pandemic i don't think the storming of the capital would have seemed likely we actually have to do the work of mapping out the space of all possible futures and this is part of planning i mean this is what good planning does this is made possible by the prefrontal cortex we can uniquely do this you know because uh we're you know at the leading edge of intelligence it's it's a practice that we have to engage in now if we want to be prepared for the future i just wanted to post this out on a little bit more of a general or personal note perhaps wanted to hear your view on how the writing and the reception of your book as opposed to say other live stream appearances or articles you've written how did that process of writing the book update your sense of science's direction especially in terms of global participation and education and research like where are we at with science inclusion participation and where is science going with all these topics that you thread together in the work like where does that bundle of science go and how to change how did writing the book update the way that you were seeing yeah so i guess the first thing i would say was that it became very apparent especially reading about evolutionary epistemology and karl popper because his work was the philosophy of science and epistemology and it became really clear that academia kind of um you know you you can't think of an idea like too radical everything you do has to be based on past research you can't invent a question and you know write a paper about like some like if you tried to submit a paper to a journal that didn't reference past work it would not have any chance of getting accepted and this is good in a way because it forces people to read the

literature and understand what's out there but it's really constraining and you have this publisher parish game that makes it such that people are just going where there's you know easy publication so uh you're and you're you're being conservative you want to just build very small incrementally on the work that has come before but the problem is that if you have this big integrated view because you've looked at all of these different fields like if you look at you know from biology to neuroscience to thermodynamics to information theory uh cybernetics um you're going to be seeing connections people that are just like biologists working in the field like and paying attention to those mainstream biology journals that they just don't see there's so so there's not enough interdisciplinary like connection collaboration um there's not enough emphasis there's not enough appreciation for generalists like everybody's like um uh basically pressured to be a specialist in science and we really need generalists right now to step back and look at all the different fields and not just fields of science but philosophy because anytime you do science you're making philosophical and metaphysical assumptions um and yeah the in the the hardest problems that you haven't understood like say the hard problem of consciousness it's always philosophy at first that's going to kind of define like the territory like make you understand like what what the problem is like how to approach it so we need generalists we need scientists working with philosophers we need scientists becoming philosophers we need philosophers caring more about science and all these mechanisms rather than just like picking pan psychism as a stance because it seems cool to you and then you just defend it for the rest of your career i also see a lot of philosophers just picking like illusionism or something and sticking to it like i think that's bad too we should all be excited about updating our model when new evidence comes in that um so yeah we we shouldn't get tied to any one idea too much and um not you know there's a role for specialists i mean people solve these really hard problems in their fields because they're just kind of have this one track mind and tunnel vision at the time we're at right now to like make progress and to really like apply this stuff like to society because societies are complex adaptive systems like it's a big mistake to think like these things can't like these scientific understanding of like evolutionary mechanisms and emergence can't inform like our institutions and our sense of morals that's a big mistake yeah writing it uh that just became very apparent that and i was lucky enough to have i mentioned before harold morowitz as a professor and he was just someone you know it was a neuroscience department but he was an origin of life researcher so you know he got us like understanding that like we we had to start thinking about like you energy like free energy and that information theory is important and epistemology um that and santa fe institute is you

know good place to mention because their work is interdisciplinary by nature we need to be that way and we need to push back on the culture that like actually some scientists are threatened by this and like the extended evolutionary synthesis like if you read that guardian article i that you know a lot of these scientists are saying that we need to extend the modern synthesis a lot of scientists were really mad and insulted by that and they feel like it's attacking this sacred thing and some of them might even you know claim that some concepts are pseudoscience i still see people thinking like emergence like some people thinking you know some of these different concepts from complexity science are like not real science and a lot of times those are people that are just have been like indoctrinated by like uh militant reductionism i won't like go like you know talk too bad about reductionism it's a great method but it's a terrible philosophy and worldview um so yeah i think there needs to be a big push to have an awareness that we need generalists we need people looking at everything and i think people who aren't in academia who are interested in this kind of stuff people who follow the channel but they're not like at a university they have a big role to play here and they can you know see these connections and blog about it whatever they do to like create content that kind of shapes the narrative because that's one thing i've seen like the narrative does get shaped by culture and by these people who are really good at articulating things um daniel schmucktenberger is kind of like a complexity theorist philosopher but you know i as far as i'm aware he doesn't have like you know he's he's not in academia but um people like this are kind of like showing us these connections and trying to create a sense-making lens out of our scientific paradigms and um i think we're going in the direction i'm i'm excited to see like i've been finding all these new communities so i'm looking for them like metamodern communities and game b and uh these people are looking for this kind of like integrated uh paradigms and it's pretty exciting thanks for the answer just to share a few more ideas and you can feel free to respond or we can go as long as you want when you were mentioning those regimes of attention which is an active inference term meaning the regimes of action and cognitive states like what entities are doing in the niche and what their cognitive models are and in the social context that relates to the concept of scripts like strong and weak scripts and when it comes to science broadly that niche and the scripts and the norms and the archetypes and all the different players include like the journals the publishers what people are paying attention to the public science meme pages like everything and then i heard the call for the normalization of transdisciplinary collaboration which implies at least like an island of shared generalism like not everyone in the transdisciplinary collaboration is going to be pan generalist but there

has to be some overlap in their venn diagram or some way that they can align in their perspectives which is related to generalism or being able to take different vistas on the differences between the perspectives um and then like that transdisciplinary whether in the mind of one person or on teams it happens and and hopefully will happen at the informal like everyday level just bringing that kind of curiosity and synthesis to like well it's really cool that the algae are growing here next to the waterfall and not here what does that mean about uncertainty or how do they how are they minimizing free energy by living here not there so that kind of like everyday embodied science conversation and then also hopefully the rigorous like decentralized science desai research that teams and communities could be carrying out i just thought it was kind of cool how this is a idea that applies to like the whole continuum what today is occupied by like research institutions doing research and then like the the citizenry and the population receiving a very certain kind of science communication and not necessarily like a including or an engaging one and then just the last point was i um heard you talk about pluralism and this sort of like bayesian pluralism like you have a portfolio of models and maybe on some features or over some time scale some fit better than another just one model in a portfolio of models train like one investing strategy or one chess move transiently doing better in a situation is hardly like something objective about that strategy or that chess move it's something that should be seen in context and across this distribution of all the chess moves and so on so i think that's a really rich angle that avenue of scientific pluralism which helen longino and others have explored is really rewarding because it just speaks to some of those questions about like modernism and game b and so on and resolving the sometimes apparent tension between like multiple truths and scientific pursuit of what is epistemically good i guess i'll just close by giving you the floor for any other comments you'd like to make and otherwise just say this was a great discussion i really appreciate the work and coming on and sharing yeah all that you said was great um totally agree uh and i think these kind of conversations just you know us recording this and putting it online and you creating this channel i think we're seeing you know all of these things are popping up partially because the pandemic it was great that like you know people had to embrace like zoom because we were going crazy um being by ourselves um but yeah i'm seeing so much excitement regarding like these kind of questions like the hard problem of consciousness like you know origin of life um all of this stuff people are super excited about to wear like all you know these people working in these fields um from science to ai they've become like celebrities and it's amazing that i i saw the kind of progression in the last few years where the the kind of the amount of

knowledge of people who weren't in uh just was like so much higher than it was before like i would say like 10 years ago being on facebook like regular people were learning about like schrodinger's cat and like making jokes about it and making memes about it and you know there were these specific ideas things that you know stephen hawking wrote about in a brief history of time like black holes like singularities whatever there are all these topics that were people were like getting into and like understanding enough to where they were becoming memes but like in the last few years i've seen this like crazy explosion where like people in these communities like these like facebook groups or on twitter i mean they know a lot they know sometimes like it seems like they have a lot more knowledge as far as like the philosophical context of one like problem than the actual like experts in the field and so i think this is great and people just need to have these dialogues they need to record them and put them online and there needs to be like this like discussion you know between different fields different groups um and yeah it's great that you talked about like you know asked about the experience writing the book um i wrote an article called is the universe pro-life and it really just argued that um life was inevitable the emergence of life it wasn't saying anything about the emergence of intelligence or anything but um so it was a pretty it was a pop article but it got into like thermodynamics on equilibrium thermodynamics and but it had this like catchy title is the universe pro-life and uh sean carroll was nice enough to give a quote for it and then retweet it with like an endorsement and i think that helped get the book deal but um so uh i was pitching it to different publishers harvard uh the harvard editor saw that article and was interested and we had lunch and i told him about it but like when he heard like the whole story and then saw the proposal he was like this is just like too big i'm not sure like which reviewers to get for this because it's just you know so many different fields it seemed like very philosophical to him it was great because then i got my uh publisher ben bella which published publishes like pop science books um and that gave me the freedom write something like this big in scope so it's great that popular science books exist i hope more scientists you know start writing like stuff like outside of their field and in journals um but it was funny because once i wrote it the you know my agent was like this is really technical i'm not sure how it's going to do because it's written at a college level and like you know usually pop books are written out like a high school level and so it turned out being a book that probably would have worked for like an academic press um but uh so yeah um i wasn't sure how it was going to be received because it had this larger kind of teleological narrative that was purely mechanistic um and uh so far it's been great it was really surprising that i got people who people like michael shermer

who are you know known for being like skeptics and atheists um because you know they felt that it was you know the the mechanisms were articulated well enough to to show that you know something like this teleological narrative was you know legitimate science and uh something really exciting but it was funny because like i won't say this person's name but they were a famous cosmologist that i thought would um be really into the ideas of the book because it were these ideas of like emergence and kind of just teleological stuff and that person was like really influential in that kind of niche area but um they said that they thought that the book was written you know super well and that they thought it would um be successful but that he couldn't endorse it because it was it talked it mentioned too many reductionists it was like too much reliant on um on richard dawkins and dan didn't it so where i thought it was really going to appeal to like the spiritual scientists it was like too reductionistic for them and then for some maybe reductionist scientists it was too much like on the emergent side so it was it became very clear there was a culture war and that we've been held back from progress just talking about like a science of agency like it an emergence these ideas for some reductionists like we're too abstract and not well defined and we're considered pseudoscience or like new age stuff or you know whatever theology both sides have to kind of see where the other side's coming from and what they have to contribute and i hope the book you know i wish i would have had a year two more years to like really make the book what it was it feels like something that's not complete or something that needs to be like continually updated because they're definitely open questions like i forgot to like put i didn't have time to like address like the fermi paradox um stuff that was like really relevant and hopefully i've talked about in these talks but um i was really happy with uh the way um people who are looking for a science book have embraced and got excited about this larger story of purpose and progress because it totally changes the narrative it really just goes from the reductionist worldview which said you know life and mind are epic phenomena and we don't have true agency and that life is transient it's a really nihilistic depressing world view to a scientific world view that does have spiritual implications we're not we're not um we're not these tren we're not destined for transients and um one person wrote me and said something like understanding this worldview made reality like our like just the general experience of being alive feel different feel like more infused with like meaning and purpose and that you know would be like one of the major goals of that is that you know people don't feel like life is some sort of fluke or cosmic accident that were actually a natural manifestation uh physical laws and this actually it's not just this optimistic picture that's fun to believe it actually gives us a

picture of where we might be going and how to deal with that so doing this thing that i mentioned about like mapping out all the counter factuals so we can ensure that we move along the trajectory that's consistent with the continual existence and progression of life so yeah let's let's open our arms to philosophy and metaphysics and anything that can inform the scientific picture because science needs philosophy amazing bobby thank you again for this and really appreciate the time and we'll plan point three when the time is right yeah so good luck with everything and see you around thank you so much daniel love the questions thanks for everybody else who is in the chat or whatever appreciate it all right thank you have a good you